Field Shift: Adapting Farms with NASA Data

NASA Space Apps Challenge 2026 | The Problem & Our Proposed Solution

1. The Problem

Farmers around the world are dealing with a changing environment: temperatures are shifting, rain is less predictable, water is becoming scarce, extreme weather is more common, and soil is losing its health (erosion and nutrient loss).

What is crop rotation?

Crop rotation means growing different crops in the same field in a planned order, season after season, instead of the same crop every time. Done well, it brings real benefits:

- **Healthier soil:** some crops (like beans) add nitrogen back to the soil; others add organic matter.
- **Less pests and disease:** changing crops breaks the life cycle of pests that depend on one plant.
- **Better water use:** deep-rooted crops help water soak in, and drought-tolerant crops use less water.
- **Less erosion:** keeping the soil covered stops wind and rain from washing it away.
- **Extra income:** adding a crop can create new products to sell, such as animal feed (forage).

Why is it hard for farmers today?

The right choice depends on the place. A crop that improves the soil in one region may use too much water, bring a disease, not sell in the local market, or not fit the growing season somewhere else. As the climate changes, old experience and old calendars become less reliable. Farmers need up-to-date, local information, but most do not have easy access to long-term satellite data or the tools to understand it.

Challenge	Why it matters
Unpredictable weather	Planting dates and crop choices based on old patterns may now fail.
Water shortage	A "good" crop may use too much water for the area.
Declining soil health	Nutrient loss and erosion reduce yields year after year.
Too much complexity	Soil, climate, pests, markets, and equipment must all be considered together.

Core problem: farmers lack clear, local, data-based guidance to choose crop rotations that protect their land and fit a changing climate.

2. Our Proposed Solution

We build a **decision-support tool** (a web app and/or interactive map) that combines **NASA Earth observation data** with **local soil information**, **crop characteristics**, and **the farmer's own priorities**. It helps the farmer explore and compare crop rotation strategies for their land, and understand why each one is suggested.

How it works (4 steps)

Step	What happens
1. Collect data	Gather NASA satellite data about the farm's location: temperature, rainfall, soil moisture, vegetation health, and water trends over many years.
2. Add local context	Combine it with local soil type, crop facts (water need, season, nitrogen-fixing ability), and the farmer's priorities (save water, extra income, improve soil...).

Step	What happens
3. Analyze patterns	Find patterns such as when fields are usually bare, when current crops are most stressed, and risks like drought, heat, or frost.
4. Compare strategies	Show at least 3 rotation options, including the farmer's current rotation, and explain each one clearly.

What the farmer sees for each rotation option

- **Why** it was suggested.
- **Evidence:** what data supports it.
- **Benefits:** for soil, water, pests, and income.
- **Information gaps:** what the tool does not know.
- **Risks:** what local farmers or agricultural experts should review.

3. NASA Data We Can Use (in simple words)

Dataset	What it tells us
SMAP	How wet or dry the soil is (soil moisture).
GPM	How much rain and snow fell.
NASA POWER	Daily weather data such as temperature and sunshine.
MODIS / Landsat	How healthy and green plants are (vegetation, e.g. NDVI).
ECOSTRESS	Land surface temperature and plant heat stress.
GRACE / GRACE-FO	Long-term changes in underground water storage.
SRTM	Land elevation and slope (helps estimate erosion and frost-prone areas).

These datasets come in different map sizes and time frequencies, so a key task is to **harmonize** them (resample them to one common grid) so they can be compared together. Satellite data is also not perfect, so combining it with ground information (local soil data and farmer knowledge) makes results more reliable.

4. Simple Example

A farmer grows wheat every year in a dry area, and their top priority is saving water. The tool compares three options: **(A) Current rotation:** wheat only; **(B) Wheat + beans:** beans add nitrogen to the soil; **(C) Wheat + a drought-tolerant crop + a cover crop:** uses less water and covers bare fields. The tool shows that satellite data finds the field bare for several months each year and soil moisture declining over time, so option C looks best for water and erosion, with a note that local experts should check the market and seed availability.

5. Key Principles

- **Supports, never replaces:** the farmer and local experts make the final decision.
- **Simple and farmer-friendly:** clear words, visuals, and maps for people with no technical background.
- **Honest about uncertainty:** shows confidence levels and where local validation is needed.
- **Farmer-centered:** the farmer can enter their own priorities and limits (water, equipment, market).

In One Sentence

We build a tool that turns NASA satellite data into clear, farmer-friendly advice, helping farmers choose the crop rotation that best protects their soil, saves water, and fits their priorities as the climate changes.